

SHUBHAM KUMAR YADAV

AI-FOCUSED FULL STACK ENGINEER · MACHINE LEARNING · CLOUD & DEVOPS

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Core skills: Python | Machine Learning | LLM and Vector Search | FastAPI | REST API | React | TypeScript | PostgreSQL | OpenSearch | Docker | Azure | AWS | CI/CD

SUMMARY

AI-focused full stack engineer with an MCA in Artificial Intelligence (CGPA 8.71), working as the sole developer on live production AI systems. Own a multi-tenant AI product-search SaaS on Azure end to end: 51 FastAPI endpoints, an OpenAI embedding and OpenSearch retrieval core, voice search, LLM agents, two React dashboards and the Docker, nginx and TLS infrastructure underneath. Restored a client's dead AWS marketplace search platform, re-indexing 255,350 products with zero errors against a frozen backend, and shipped a nightly scikit-learn risk model at 0.936 ROC AUC. Operate production systems solo, write the runbooks and work directly with clients. Open to relocation and to remote roles.

EXPERIENCE

ATH Infosystems, Noida — Full Stack Engineer (AI/ML, DevOps & Cloud)

Dec 2025 – Present

- Own the company's commercial multi-tenant AI product-search SaaS as sole developer: 51 REST API endpoints across ingestion, search, analytics, billing, admin and client-portal services in Python and FastAPI, two React dashboards, and the production stack behind them — Docker Compose running OpenSearch, PostgreSQL and Redis, with FastAPI behind nginx and Let's Encrypt TLS on an Azure VM I provisioned.
- Engineer the semantic search core with OpenAI text-embedding-3-large (3072-dimension vectors) indexed into OpenSearch, so plain-language queries such as "gift for someone who cooks" return relevant cookware instead of keyword matches; cut search latency from ~500 ms to ~150-250 ms (repeat queries ~0 ms) with a two-tier embedding cache, single-pass query execution and off-event-loop search calls, and shrank responses from 83 KB to 19 KB with gzip.
- Shipped **voice search** end to end: a Web Speech API microphone in the storefront widget and both dashboards, an API returning results plus an LLM spoken summary read back by text-to-speech, with voice AI cost folded into per-request billing and a by-source analytics tab.
- Ship an in-product AI helpdesk agent on an OpenAI tool-calling loop with three production tools (add synonym, remove synonym, raise support ticket), where prompt and tool design enforce privacy walls so no tenant can reach another tenant's data through the agent.
- Built conversational query understanding: queries of six or more words go through an LLM intent step extracting core product, price limits and attributes (cached 24 h, fail-open), so stated price and stock limits survive every search fallback.
- Enforce multi-tenancy at the API layer by scoping every call by client_id behind dynamic per-tenant isolation guards, verified by a 31-test automated tenant-isolation suite.
- Restored a large e-commerce marketplace client's dead AWS-based search platform onto Azure under a strictly frozen backend, re-indexing 255,350 products with zero indexing errors in 1 hour 52 minutes.
- Built a nightly machine learning withdrawal-risk model for an education-sector client in Python, PySpark and scikit-learn (logistic regression) reaching 0.936 ROC AUC and scoring 1,300+ students into a Delta Lakehouse, and authored 55+ DAX measures wired into a natural-language AI assistant so business users get answers computed live from governed measures.
- Took over sole developer ownership of a US FinTech client's live production platform after the original developer left, and now own its codebase, AWS infrastructure, backups, releases and client-facing technical responses.
- Coordinated a small project team, mentored one junior developer and ran client communication from requirements through demos and support; wrote the platform's 51-page technical handbook, client integration guides, portal user guides and deployment runbooks.

Salesqueen Software Solutions, Chennai — Front-End Developer (Internship)

Mar 2025 – Jul 2025

Remote, alongside final-semester MCA

- Built responsive, accessible UI components across six delivered products, including a login portal, production portal, Kanban board, e-commerce website and an online cab-booking app.
- Deployed and maintained releases on live servers and kept the interface layer consistent across the product suite.

CSRBOX Foundation via IBM SkillsBuild — AI Intern (Remote)

Jul 2024 – Aug 2024

- Built a conversational chatbot using IBM Watsonx AI and NLP techniques.

Gautam Buddha University, Greater Noida — Admission Department Intern

Jun 2024 – Aug 2024

- Managed a 3,000+ student record dataset in MySQL plus CRM data through the admission cycle, and built an online Bonafide Certificate system that issued certificates from those records.

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript (ES6+), TypeScript, SQL, DAX, Bash, PHP, Kotlin, HTML5, CSS3
- **AI / Machine Learning & Search:** machine learning, OpenAI embeddings, Large Language Models (LLM), LLM tool-calling, prompt & tool design, semantic and vector search, OpenSearch, RAG (Retrieval-Augmented Generation), scikit-learn, PySpark, IBM Watsonx AI, NLP
- **Backend & APIs:** FastAPI, Node.js, Express.js, REST API design, GraphQL, JWT, OAuth2, webhooks, multi-tenant API architecture, Stripe API, PayPal API, Microsoft Graph API, SCIM
- **Frontend:** React, Vite, Tailwind CSS, Bootstrap 5, Recharts, Chart.js, responsive and accessible UI
- **Cloud & DevOps:** Azure (VMs, Logic Apps, Key Vault, Entra ID, DevOps), AWS (EC2, ECS Fargate, RDS, S3, CloudFront, IAM, OpenSearch Serverless, ElastiCache, Secrets Manager), Terraform, Docker, Docker Compose, Ansible, GitHub Actions, CI/CD, nginx, TLS/certbot, Linux (Ubuntu)
- **Databases:** PostgreSQL, MySQL, MongoDB (Atlas), Redis, SQLite, Firebase
- **BI & Data:** Microsoft Fabric, Power BI, DAX, Tabular Editor, Delta Lake / OneLake, Copilot Studio
- **Tools & Practices:** Git, GitHub, Postman, Figma, technical documentation & runbooks, client communication, mentoring

KEY PROJECTS

AI Lead-Capture Assistant

ATH Infosystems

Embeddable chat widget live on two company websites

Built a Flask chat service and embeddable JavaScript widget live on the company's two public sites: answers visitor questions through an LLM and captures qualified leads straight into SQL Server, with a lead admin console, CSV export and Swagger-documented APIs. Authored 44 of the project's 60 commits.

Legacy Marketplace Search Platform — Restore & Modernisation

Large e-commerce marketplace client · ATH Infosystems

Engineered a local TLS/SigV4 compatibility shim so backend code written for AWS OpenSearch Serverless ran unmodified against self-hosted OpenSearch, meeting a zero-code-change constraint on the client's API. Built a FastAPI service powering two new React dashboards behind nginx, wrote idempotent bash installers that redeploy the whole stack with one command, and fixed a silent analytics event-loss bug with a schema repair script.

FinTech Income-Splitting SaaS

US client · live production · inherited ownership · ATH Infosystems

Own a live React, TypeScript, Express and PostgreSQL platform on AWS (RDS, ECS Fargate, CloudFront) taken over mid-project. Ran a security audit documenting critical findings — an unauthenticated payment endpoint, an IDOR, sensitive data in logs and secrets committed to environment files — and produced the remediation plan. Found a revenue-affecting gap where no PayPal webhook was registered, so subscriptions expired silently while customers kept being billed. Executed full production backups ahead of a controlled infrastructure pause. Domain: Stripe Connect / Financial Connections, ACH debits, automated payouts, PayPal Billing.

AI-Powered Predictive Analytics & Reporting Platform

Education-sector client · ATH Infosystems

Microsoft Fabric and Power BI platform for enrolment forecasting, retention and admissions: layered semantic models, cross-source CRM-to-SIS integration with row-level security, and a natural-language AI assistant answering plain-English questions from the governed DAX measures.

Automated HR-to-Cloud Identity Provisioning Pipeline

ATH Infosystems

Azure Logic Apps pipeline syncing BambooHR to Microsoft Entra ID via the Microsoft Graph API and the SCIM bulkUpload endpoint, with OAuth2, Azure Key Vault, duplicate-conflict detection, automated group assignment and templated Office 365 welcome emails — a self-healing idempotent design that removed manual onboarding entirely.

Personal Projects

- **To-Do App with automated CI/CD** ([GitHub](#)) — deployed across 11 AWS EC2 instances via Ansible triggered by GitHub Actions. **Expense Tracker** ([Live Demo](#)) — Node.js, GraphQL, JWT, React. **Android chat app** ([GitHub](#)) — Kotlin, Firebase.

EDUCATION

Gautam Buddha University — MCA (Artificial Intelligence), CGPA 8.71

Jul 2023 – Jul 2025

Veer Narmad South Gujarat University — BCA, CGPA 7.61

Oct 2020 – Jun 2023

CERTIFICATIONS

- DevOps and Cloud Computing — PW Skills · [Certificate](#) · AI Internship, Watsonx AI — IBM SkillsBuild / CSRBOX · [Certificate](#)
- Front-End Developer — Salesqueen · [Certificate](#) · Internship — GBU Admission Department · [Certificate](#)

Languages: English, Hindi